Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction

Period descriptors, PP-kernel nodes,
and the full tractability of RCC5

(Revised, April 2026)

Claude (Anthropic)
Prompted by Michael Wessel

April 2026

Disclaimer. This document was generated by Claude (Anthropic), prompted by Michael Wessel. The results have not been independently verified. We invite scrutiny and corrections.

Revision note. This version addresses five technical objections raised by GPT-5.4 Pro [5] in a review of the original manuscript [3]. The three substantive fixes are: (1) designated witnesses indexed by $\mathsf{Union}(\mathsf{Desc})$ instead of $\mathsf{Core}(\mathsf{Desc})$; (2) a type-safety filter Safe added to V6; (3) kernel nodes indexed by (descriptor, external interface) pairs. Two further clarifications concern the certificate nature of the quotient and the blocking invariant.

Abstract

We prove that concept satisfiability in $\mathcal{ALCI}_{RCC5}$—the description logic $\mathcal{ALCI}$ extended with an RCC5 role box—is decidable. The proof proceeds by a model-theoretic quotient construction. Every model of a satisfiable concept can be represented by a finite *two-tier quotient*: **Tier 1** captures each infinite PP-chain's eventually periodic tail as a finite *period descriptor* (a cyclic word of Hintikka types), while **Tier 2** captures cross-chain interactions through *PP-kernel nodes* indexed by (descriptor, external interface) pairs. Conversely, every valid two-tier quotient can be unfolded into a genuine model, using the *full tractability* of RCC5 (Renz & Nebel 1999): path-consistent disjunctive RCC5 networks are satisfiable. The off-chain extension condition (V6) incorporates both RCC5 composition constraints and a *type-safety filter* ensuring respect of universal restrictions. Together, these yield a decision procedure: enumerate quotients up to a computable size bound and check validity. The key algebraic ingredient is that PP is *universally self-absorbing*, ensuring composition consistency of the kernel representation.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with five jointly exhaustive and pairwise disjoint (JEPD) base relations from the Region Connection Calculus RCC5 [10]: DR, PO, EQ, PP, PPI. Wessel [1] introduced this logic and left its decidability open. The logic lacks both the finite model property and the tree model property: the concept $C_\infty = (\exists\mathsf{PP}.\top) \sqcap (\forall\mathsf{PP}.\exists\mathsf{PP}.\top)$ is satisfiable only in infinite models containing unbounded PP-chains.

Several recent approaches have been attempted: a quasimodel method [2] (gap in completeness), a contextual tableau [4] (whose completeness conjecture $\mathsf{FW}(C, N)$ is false [6]), and a triangle-type

approach [7] (conditional on an unproven conjecture). All share a common difficulty: assigning globally consistent RCC5 edges in complete-graph models.

In this paper we resolve the decidability question affirmatively. The proof has two parts:

Completeness (satisfiable concept $\Rightarrow$ finite quotient): Given any model $\mathcal{I} \models C_0$, we extract a finite *two-tier quotient*. Each infinite PP-chain is represented by a *period descriptor*—a cyclic word $(\tau_1, \ldots, \tau_p)$ of Hintikka types capturing the chain's eventually periodic structure. Cross-chain interactions are represented by *PP-kernel nodes* indexed by (descriptor, external interface) pairs. The model's global consistency guarantees that the quotient satisfies a checkable *extension consistency* condition (V6), which incorporates both RCC5 composition constraints and a type-safety filter.

Soundness (valid quotient $\Rightarrow$ satisfiable concept): Given a valid two-tier quotient, we unfold it into a model. The quotient is an abstract *certificate*, not itself an RCC5 model: kernel nodes carry reflexive PP-loops as a bookkeeping device, which are replaced by genuine EQ diagonals during chain unfolding. Period descriptors become infinite PP-chains. Off-chain witnesses are added via an extension procedure whose consistency is guaranteed by V6 and the *full tractability of RCC5* [8]: path-consistent disjunctive RCC5 constraint networks are satisfiable.

The decision procedure enumerates all two-tier quotients up to a computable size bound and checks validity. The quotient size is bounded by a computable (though possibly non-elementary) function of the input concept.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts are built from concept names, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$ in negation normal form (NNF). We adopt **strong EQ semantics**.

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ assigns a total function $\rho^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ subject to: (i) $\rho(d, d) = \mathsf{EQ}$; (ii) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$; (iii) composition consistency: $\rho(d_1, d_3) \in \mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$ for all d_1, d_2, d_3.

2.2 Types, closure, composition table

Given C_0 in NNF, $\mathsf{cl}(C_0)$ is the closure under subformulas. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset. Write $\mathsf{Tp}(C_0)$ for the set of types; $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

The RCC5 composition table $\mathsf{comp} : \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}^2 \to 2^{\{\mathsf{DR},\mathsf{PO},\mathsf{PP},\mathsf{PPI}\}}$ (with EQ excluded for distinct elements) is:

$\mathsf{comp}(R_1, R_2)$	DR	PO	PP	PPI
DR	{all}	{DR,PO,PP}	{DR,PO,PP}	{DR}
PO	{DR,PO,PPI}	{all}	{PO,PP}	{DR,PO,PPI}
PP	{DR}	{DR,PO,PP}	{PP}	{all}
PPI	{DR,PO,PPI}	{PO,PPI}	{PO,PP,PPI}	{PPI}

2.3 Type-safety for RCC5 relations

Definition 2.1 (Type-safe relations)**.** For Hintikka types $\tau_1, \tau_2 \in \mathsf{Tp}(C_0)$, define

$$\mathsf{Safe}(\tau_1, \tau_2) = \{R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} : R \text{ respects all universals in } \tau_1 \text{ and } \tau_2\}$$

where R *respects the universals* means: for every $\forall R.C \in \tau_1$, $C \in \tau_2$; and for every $\forall \mathsf{inv}(R).C \in \tau_2$, $C \in \tau_1$.

Lemma 2.2 (Model relations are type-safe)**.** *In any model $\mathcal{I}$, for distinct $d, e \in \Delta^{\mathcal{I}}$: $\rho(d, e) \in \mathsf{Safe}(\mathsf{tp}(d), \mathsf{tp}(e))$.*

Proof. Immediate from the semantics: if $\forall R.C \in \mathsf{tp}(d)$ and $\rho(d, e) = R$, then $e \in C^{\mathcal{I}}$, so $C \in \mathsf{tp}(e)$. The inverse direction is symmetric. □

2.4 Full tractability of RCC5

The following deep result from qualitative spatial reasoning is central to our proof.

Theorem 2.3 (Full RCC5 tractability, Renz & Nebel [8], Renz [9])**.** *Every path-consistent disjunctive RCC5 constraint network over the base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ is satisfiable. Equivalently: if path-consistency enforcement (= arc-consistency for the composition constraints) does not empty any domain, then a consistent atomic refinement exists.* □

This is the "patchwork property" of RCC5: local consistency (path-consistency) implies global consistency. For atomic networks: every path-consistent atomic RCC5 network is realizable by regions in $\mathbb{R}^n$ for sufficiently large n.

3 Algebraic properties of PP

Theorem 3.1 (Universal self-absorption of PP)**.** *(a)* ***Left absorption:*** *$R \in \mathsf{comp}(\mathsf{PP}, R)$ for every $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.*

(b) ***Right absorption:*** *$R \in \mathsf{comp}(R, \mathsf{PP})$ for every R.*

(c) ***Self-closure:*** *$\mathsf{PP} \in \mathsf{comp}(R, \mathsf{inv}(R))$ for every R.*

(d) ***PPI right-absorption:*** *$R \in \mathsf{comp}(\mathsf{PPI}, R)$ for every R.*

(e) ***PPI left-absorption:*** *$R \in \mathsf{comp}(R, \mathsf{PPI})$ for every R.*

Proof. Direct verification against the composition table. Parts (d)–(e) are the dual of (a)–(b) under the involution $\mathsf{PP} \leftrightarrow \mathsf{PPI}$. □

Corollary 3.2 (Reflexive PP is composition-consistent)**.** *A node k with $\rho(k, k) = \mathsf{PP}$ satisfies all composition constraints with any external node m.* □

Theorem 3.3 (PP-transitivity and strict order)**.** $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$. *Consequently,* PP *on any finite set of elements is a strict partial order, and no* PP*-cycle exists among distinct elements.* □

Lemma 3.4 (Non-empty compositions)**.** $\mathsf{comp}(R_1, R_2) \neq \emptyset$ *for all $R_1, R_2 \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.* □

4 PP-chain structure

Definition 4.1 (PP-chain)**.** In a model $\mathcal{I}$, a PP-*chain* is a sequence $(d_i)_{i\geq 0}$ with $\rho(d_i, d_{i+1}) = \mathsf{PP}$ for all i. By Theorem 3.3, $\rho(d_i, d_j) = \mathsf{PP}$ for $i < j$.

Lemma 4.2 (Monotone accumulation)**.** *Define* $F_i = \{C \in \mathsf{cl}(C_0) : \forall\mathsf{PP}.C \in \mathsf{tp}(d_i)\}$. *Then* $F_0 \subseteq F_1 \subseteq \cdots$, *and the sequence stabilizes within* $|\mathsf{cl}(C_0)|$ *steps.* □

Definition 4.3 (Stabilized tail)**.** The *stabilized tail* of a PP-chain $(d_i)_{i\geq 0}$ is $(d_i)_{i\geq N}$ where N is the stabilization index.

Lemma 4.4 (External relation stabilization)**.** *For external element* e *and chain* (d_i), *the sequence* $R_i = \rho(e, d_i)$ *satisfies* $R_{i+1} \in \mathsf{comp}(R_i, \mathsf{PP})$ *and stabilizes.* PP *is the unique absorbing value:* $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$. □

Definition 4.5 (Recurrent types, T_∞)**.** For a PP-chain $(d_i)_{i\geq 0}$, define $T_\infty = \{\tau \in \mathsf{Tp}(C_0) : \tau = \mathsf{tp}(d_i)$ for infinitely many $i\}$.

Lemma 4.6 (T_∞ is non-empty and eventually exhaustive)**.** $T_\infty \neq \emptyset$. *There exists* $M \geq N$ *(the stabilization index) such that* $\mathsf{tp}(d_i) \in T_\infty$ *for all* $i \geq M$.

Proof. Since $|\mathsf{Tp}(C_0)|$ is finite and the sequence $(\mathsf{tp}(d_i))_{i\geq 0}$ is infinite, some type appears infinitely often (pigeonhole), so $T_\infty \neq \emptyset$. Types not in T_∞ appear only finitely often; let M be the last index at which any non-T_∞ type appears. For $i > M$, $\mathsf{tp}(d_i) \in T_\infty$. □

5 Period descriptors

Definition 5.1 (Period descriptor)**.** A *period descriptor* for C_0 is a tuple $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ with $p \geq 1$ and each $\tau_j \in \mathsf{Tp}(C_0)$, satisfying:

(V1) Each τ_j is a Hintikka type in $\mathsf{cl}(C_0)$.

(V2) **∀PP-closure:** $\forall\mathsf{PP}.C \in \tau_i$ for some i implies $C \in \tau_j$ for all j.

(V3) **∀PPI-closure:** $\forall\mathsf{PPI}.C \in \tau_i$ for some i implies $C \in \tau_j$ for all j.

(V4) **Chain continuation:** For each i, there exists $\exists\mathsf{PP}.C \in \tau_i$ such that $C \in \tau_j$ for some j.

The descriptor represents the infinite periodic chain $\cdots \tau_1\ \mathsf{PP}\ \tau_2\ \mathsf{PP}\ \cdots\ \mathsf{PP}\ \tau_p\ \mathsf{PP}\ \tau_1\ \mathsf{PP}\ \cdots$

Remark 5.2 (V4 is chain continuation, not full witness closure)**.** Condition V4 requires that each type has *at least one* ∃PP demand satisfiable within the period (ensuring the chain can continue). Other ∃PP demands of τ_i may be satisfied by off-chain witnesses (handled by Tier 2 of the quotient). This weakening is essential: an element on a PP-chain may have multiple ∃PP demands, only one of which generates the next chain element. The others generate off-chain PP-branches.

Theorem 5.3 (T_∞ yields a valid period descriptor)**.** *Let* $(d_i)_{i\geq 0}$ *be a* PP-*chain in a model* $\mathcal{I} \models C_0$. *Any enumeration of* $T_\infty = \{\tau_1, \ldots, \tau_p\}$ *is a valid period descriptor.*

Proof. (V1): Each $\tau_j \in T_\infty \subseteq \mathsf{Tp}(C_0)$.

(V2): If $\forall\mathsf{PP}.C \in \tau_i$ for $\tau_i \in T_\infty$, then τ_i appears at some position $k \geq M$ on the chain. Since $\forall\mathsf{PP}.C \in \mathsf{tp}(d_k)$ and $\rho(d_k, d_j) = \mathsf{PP}$ for all $j > k$, we have $C \in \mathsf{tp}(d_j)$ for all $j > k$. In particular, $C \in \tau$ for every $\tau \in T_\infty$ (since every $\tau \in T_\infty$ appears at some position $j > k$).

(V3): Dual argument using $\forall\mathsf{PPI}$ and backward propagation. If $\forall\mathsf{PPI}.C \in \tau_i$ at position k, then $C \in \mathsf{tp}(d_j)$ for all $j < k$ with $\rho(d_k, d_j) = \mathsf{PPI}$. On the chain, $\rho(d_k, d_j) = \mathsf{PPI}$ for $j < k$. So C is in all types at positions $< k$. Since τ_i appears at arbitrarily large k (infinitely often), C is in all types at all positions $> M$, hence in all $\tau \in T_\infty$.

(V4): Each d_k (with $k \geq M$, $\mathsf{tp}(d_k) \in T_\infty$) has a PP-successor d_{k+1} on the chain. The element d_{k+1} witnesses some $\exists\mathsf{PP}.C$ demand of d_k, and $\mathsf{tp}(d_{k+1}) \in T_\infty$. So $C \in \mathsf{tp}(d_{k+1})$ for some $\mathsf{tp}(d_{k+1}) \in T_\infty$. □

Definition 5.4 (Stabilized core and union)**.** For $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$:

- $\mathsf{Core}(\mathsf{Desc}) = \bigcap_{j=1}^{p} \tau_j$. By (V2) and (V3), $\mathsf{Core}(\mathsf{Desc})$ contains all $\forall\mathsf{PP}$ and $\forall\mathsf{PPI}$ consequences.
- $\mathsf{Union}(\mathsf{Desc}) = \bigcup_{j=1}^{p} \tau_j$. This is the set of *all* formulas occurring in *any* phase type of the descriptor.

6 The two-tier quotient

6.1 Witness trees and PP-chains

Fix a model $\mathcal{I} \models C_0$. The *tree unraveling* $\mathcal{I}^T$ of $\mathcal{I}$ is a tree-shaped model where each node is a domain element and each edge corresponds to an existential witness. In $\mathcal{I}^T$:

- The root satisfies C_0.
- Each node d has one child for each $\exists R.C \in \mathsf{tp}(d)$ (the child witnesses the demand via relation R).
- A *maximal* PP*-chain* is a maximal path through the tree following $\exists\mathsf{PP}$-witness edges.

Remark 6.1. A node may have multiple $\exists\mathsf{PP}$ demands. One generates the next element on the main PP-chain; the others generate *off-chain* PP*-branches*, which are themselves roots of sub-PP-chains. The tree of PP-chains forms a recursive structure where each chain may spawn sub-chains at each position.

6.2 Definition of the two-tier quotient

Definition 6.2 (Two-tier quotient)**.** A *two-tier quotient* $\mathcal{Q}$ for C_0 consists of:

Tier 1 (within-chain structure): A finite set of period descriptors $\{\mathsf{Desc}_1, \ldots, \mathsf{Desc}_K\}$, each satisfying (V1)–(V4).

Tier 2 (cross-chain structure):

(T1) A set of *PP-kernel nodes* $\{k_1, \ldots, k_K\}$, one per (descriptor, external interface) pair. Each k_α is associated with a descriptor Desc_α and an *external interface* σ_α (the function mapping every other quotient node to a base relation). We set $\mathsf{tp}(k_\alpha) = \mathsf{Core}(\mathsf{Desc}_\alpha)$ and $\rho(k_\alpha, k_\alpha) = \mathsf{PP}$ (as a bookkeeping device; see Remark 6.3).

(T2) A set of *regular nodes* $\{m_1, \ldots, m_L\}$ with types $\mathsf{tp}(m_\beta) \in \mathsf{Tp}(C_0)$.

(T3) An atomic RCC5 labeling ρ on all pairs of kernel and regular nodes (excluding self-pairs of regular nodes, which have $\rho = \mathsf{EQ}$), satisfying composition consistency for every triple.

(T4) For each kernel node k_α and each non-PP existential demand $\exists R.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ with $R \neq \mathsf{PP}$: a designated witness among the kernel/regular nodes, such that the designated witness w satisfies $C \in \mathsf{tp}(w)$, $\rho(k_\alpha, w) = R$, and $R \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$ for every phase type τ_j containing $\exists R.C$.

(T5) For each regular node m_β and each existential demand $\exists R.C \in \mathsf{tp}(m_\beta)$: a designated witness.

Extension consistency:

(V5) **Cross-chain coherence:** The atomic network on kernel and regular nodes is composition-consistent.

(V6) **Off-chain extension:** For each $\exists \mathsf{PP}.C$ demand of a kernel node k_α that is *not* satisfied within Desc_α (i.e., $C \notin \tau_j$ for any j), there is a witness node w with $\rho(k_\alpha, w) = \mathsf{PP}$ and $C \in \mathsf{tp}(w)$. Furthermore, the disjunctive constraint network obtained by adding all such off-chain witnesses simultaneously (with demanded relations to their parents and initial domains

$$D(w, e) = \mathsf{comp}(\text{demanded}, \rho(\text{parent}, e)) \cap \mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e)) \tag{1}$$

for all existing nodes e) is *path-consistent* after arc-consistency enforcement (i.e., no domain becomes empty).

Remark 6.3 (The quotient is a certificate, not a model)**.** The two-tier quotient is *not* itself an RCC5 network in the strong-EQ sense. In particular, the reflexive PP-loops on kernel nodes violate $\rho(d, d) = \mathsf{EQ}$. The quotient is an *abstract certificate*: a finite data structure from which a genuine strong-EQ model can be constructed (Theorem 8.1). The PP self-loop is a bookkeeping device that ensures kernel-level composition checks correctly mirror within-chain behavior, via the self-absorption of PP (Theorem 3.1). During chain unfolding, each kernel is replaced by infinitely many distinct elements with EQ self-loops. Full RCC5 tractability (Theorem 2.3) is invoked only on the resulting network where all diagonal entries are EQ.

Remark 6.4 (Union vs. Core for designated witnesses)**.** The original version of this paper required designated witnesses only for demands in $\mathsf{Core}(\mathsf{Desc}_\alpha) = \bigcap_j \tau_j$. As observed by GPT-5.4 Pro [5], this is too restrictive: a demand $\exists \mathsf{DR}.D \in \tau_A \setminus \tau_B$ would lack a witness. The revised condition (T2, item 4) requires witnesses for all non-PP demands in $\mathsf{Union}(\mathsf{Desc}_\alpha) = \bigcup_j \tau_j$, ensuring that every chain element has a designated witness for every non-PP demand in its type.

Remark 6.5 (Type-safety filter in V6)**.** The original version of V6 used $D(w, e) = \mathsf{comp}(\text{demanded}, \rho(\text{parent}, e))$ without a type-safety filter. As observed by GPT-5.4 Pro [5], this permits RCC5-consistent but DL-inconsistent assignments (e.g., $\mathsf{PO}(w, e)$ when $\forall \mathsf{PO}.\neg A \in \mathsf{tp}(e)$ and $A \in \mathsf{tp}(w)$). The revised V6 intersects with $\mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$. Completeness is preserved because model-derived relations are always type-safe (Lemma 2.2). The intersection may reduce singleton composition entries ($\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$, $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$) to the empty set in principle, but this cannot happen in the completeness direction: the model's relation is the unique composition-forced value *and* is type-safe (the model satisfies all universals).

Remark 6.6. The off-chain witnesses in V6 may themselves have existential demands that require further witnesses, forming a tree of extensions. V6 checks the *joint* path-consistency of all such extensions up to a blocking depth (see Section 7). This is a finite computation.

7 Completeness: model implies quotient

Theorem 7.1 (Quotient extraction)**.** *If C_0 is satisfiable, there exists a valid two-tier quotient $\mathcal{Q}$ for C_0 of size bounded by a computable function $f(|C_0|)$.*

Proof. Let $\mathcal{I} \models C_0$. Form the tree unraveling $\mathcal{I}^T$.

Step 1: Period descriptors. For each maximal PP-chain in $\mathcal{I}^T$, compute T_∞ (Definition 4.5). By Theorem 5.3, any enumeration of T_∞ is a valid period descriptor satisfying (V1)–(V4). The number of distinct descriptors is at most $2^{|\mathsf{Tp}(C_0)|} \leq 2^{2^{|\mathsf{cl}(C_0)|}}$ (bounded by the number of subsets of $\mathsf{Tp}(C_0)$).

Step 2: Kernel nodes and stabilized relations. For each chain, compute the descriptor Desc_α and the *external interface* σ_α: for every other chain's kernel and every regular node, the doubly stabilized relation. Create one kernel node k_α per distinct $(\mathsf{Desc}_\alpha, \sigma_\alpha)$ pair, with $\mathsf{tp}(k_\alpha) = \mathsf{Core}(\mathsf{Desc}_\alpha)$ and $\rho(k_\alpha, k_\alpha) = \mathsf{PP}$.

For two kernel nodes k_α, k_β: $\rho(k_\alpha, k_\beta)$ is the doubly stabilized relation $\rho(a_i, b_j)$ for sufficiently large i, j. This stabilizes by Lemma 4.4 applied twice (first fix j, stabilize in i; then stabilize the result in j).

Two chains with the same descriptor but different stabilized external relations receive distinct kernel nodes. The number of distinct (Desc, σ) pairs is bounded: at most D descriptors, and for each, at most 4^{K+L} external interfaces, where K is the kernel count and L the regular node count. Since K and L are themselves bounded (see Step 4), this is a computable bound.

Step 3: Composition consistency of the kernel network. The kernel-level relations are stabilized model relations. For any triple $(k_\alpha, k_\beta, k_\gamma)$: choose i, j, ℓ large enough that all three pairwise relations have stabilized. The triple (a_i, b_j, c_ℓ) in the model is composition-consistent (the model is globally consistent). The stabilized relations equal the model relations at these indices, hence (V5) holds.

Similarly for triples involving kernel nodes and regular nodes (regular nodes have their original model relations).

Step 4: Designated witnesses for kernel demands. For each $\exists R.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ with $R \neq \mathsf{PP}$: since every type in T_∞ appears infinitely often on the stabilized tail, there exist chain elements of each type τ_j at arbitrarily large indices. Each such element has its demand $\exists R.C$ (when $\exists R.C \in \tau_j$) witnessed by some element w in the model with $C \in \mathsf{tp}(w)$ and $\rho(\text{chain element}, w) = R$. The model satisfies $\forall R.C' \in \tau_j$ (ensuring $C' \in \mathsf{tp}(w)$) and $\forall \mathsf{inv}(R).C' \in \mathsf{tp}(w)$ (ensuring $C' \in \tau_j$), so $R \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$. We include w as a regular node (or identify it with an existing designated witness).

Step 5: Regular nodes and blocking. The finite prefix of each PP-chain (before the stabilized tail) and all off-chain witnesses (up to a blocking depth) are recorded as regular nodes. Two off-chain witnesses w, w' are *blocked* (identified) if they share:

- the same Hintikka type τ,
- the same demanded relation R to their parent,
- the same stabilized relations to all kernel nodes.

The number of blocking classes is at most $|\mathsf{Tp}(C_0)| \times 4 \times 4^K$ where K is the number of kernels. The tree of off-chain witnesses has depth bounded by the number of blocking classes (when a class

repeats on a branch, the subtree below the repeat can be redirected to the earlier occurrence). So the number of regular nodes is bounded by a computable function of $|C_0|$.

Claim 7.2 (Blocking invariant)**.** *If w, w' share the same blocking key, then for every existing node e in the quotient, $D(w, e) = D(w', e)$ (the V6 domains are identical).*

Proof of Claim. By (1), $D(w, e) = \mathsf{comp}(R_{\text{dem}}, \rho(\text{parent}, e)) \cap \mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$. The demanded relation R_{dem} is part of the blocking key. The composition $\mathsf{comp}(R_{\text{dem}}, \rho(\text{parent}, e))$ depends on R_{dem} and the parent-to-e relation; when e is a kernel node, $\rho(\text{parent}, e)$ is determined by the parent's kernel relations (part of the blocking key for the parent). The type-safety filter $\mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$ depends only on $\mathsf{tp}(w)$ (part of the blocking key) and $\mathsf{tp}(e)$. So blocked witnesses have identical domains. □

By the blocking invariant, identifying witnesses with the same blocking key preserves V6: the disjunctive network after identification has the same domain constraints as the original, so if the original was AC-consistent, the identified version is too.

Step 6: Extension consistency (V6). Each off-chain witness w in the model has actual RCC5 relations to all other elements. We show the model's assignment survives arc-consistency enforcement on the type-safety-filtered domains.

For any existing element e in the quotient, the model gives $\rho^{\mathcal{I}}(w, e)$. This value satisfies:

(i) $\rho(w, e) \in \mathsf{comp}(\rho(w, \text{parent}), \rho(\text{parent}, e))$ (composition in the model);

(ii) $\rho(w, e) \in \mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$ (the model respects all universal constraints, by Lemma 2.2);

(iii) $\rho(w, e) \in \mathsf{comp}(\rho(w, e'), \rho(e', e))$ for every other element e' (global consistency of the model).

By (i) and (ii), $\rho(w, e) \in D(w, e)$ (the initial domain). Arc-consistency enforcement removes a value r from domain $D(w, e)$ only if there exists e' such that $\mathsf{comp}(r, \rho(e, e')) \cap D(w, e') = \emptyset$. But $\rho(w, e')$ is in $D(w, e')$ (by the same argument) and $\rho(w, e') \in \mathsf{comp}(\rho(w, e), \rho(e, e'))$ (global consistency), so $\mathsf{comp}(\rho(w, e), \rho(e, e')) \cap D(w, e') \ni \rho(w, e') \neq \emptyset$. Therefore $\rho(w, e)$ is never removed. All domains remain non-empty. □

Remark 7.3 (The critical step)**.** The proof of V6 (Step 6) is where the global consistency of the model is used most essentially. The model provides a specific atomic assignment for each off-chain witness that is simultaneously consistent with *all* composition constraints *and* all type-safety constraints. Arc-consistency enforcement cannot eliminate a value that participates in a globally consistent, type-safe assignment. This is the key insight that closes the "extension gap" identified in earlier approaches.

8 Soundness: quotient implies model

Theorem 8.1 (Quotient realization)**.** *If a valid two-tier quotient $\mathcal{Q}$ exists for C_0 (i.e., $\mathcal{Q}$ satisfies V1–V6), then C_0 is satisfiable.*

Proof. We construct a model $\mathcal{I}' \models C_0$ from $\mathcal{Q}$.

Step 1: Unfold period descriptors into chains. For each descriptor $\mathsf{Desc}_\alpha = (\tau_1, \ldots, \tau_p)$, create an infinite PP-chain $d_1^{(\alpha)}, d_2^{(\alpha)}, d_3^{(\alpha)}, \ldots$ with $\mathsf{tp}(d_i^{(\alpha)}) = \tau_{((i-1) \bmod p)+1}$ and $\rho(d_i^{(\alpha)}, d_j^{(\alpha)}) = \mathsf{PP}$ for

$i < j$. Crucially, each $d_i^{(\alpha)}$ is a distinct domain element with $\rho(d_i^{(\alpha)}, d_i^{(\alpha)}) = \mathsf{EQ}$ (strong-EQ semantics). The kernel's reflexive PP is *not* inherited by individual chain elements; it served only to validate composition constraints at the abstract certificate level.

Step 2: Assign inter-chain and chain-to-regular relations. For chain elements $d_i^{(\alpha)}$ (on the periodic tail, i large enough) and any other node e: set $\rho(d_i^{(\alpha)}, e) = \rho(k_\alpha, e)$, the kernel's relation.

This is well-defined because the kernel's relation is the stabilized value. For chain elements in the finite prefix, use the regular nodes' recorded relations.

Step 3: Verify within-chain consistency. For a triple (d_i, d_j, d_k) on the same chain with $i < j < k$: $\rho(d_i, d_j) = \rho(d_j, d_k) = \mathsf{PP}$ and $\rho(d_i, d_k) \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$. ✓

For a triple (d_i, d_j, e) with d_i, d_j on the same chain and e external: $\rho(d_i, d_j) = \mathsf{PP}$, $\rho(d_i, e) = \rho(d_j, e) = R$ (both stabilized to R). Need: $R \in \mathsf{comp}(\mathsf{PPI}, R)$. This holds by Theorem 3.1(d). ✓

For a triple (d_i, e_1, e_2) with d_i on a chain and e_1, e_2 external: $\rho(d_i, e_1) = \rho(k_\alpha, e_1)$, $\rho(d_i, e_2) = \rho(k_\alpha, e_2)$, $\rho(e_1, e_2)$ from the quotient. This is exactly the triple (k_α, e_1, e_2), which is composition-consistent by (V5). ✓

Step 4: Add off-chain witnesses. V6 ensures: the disjunctive network obtained by adding all off-chain witness variables (with their demanded relations and initial domains from (1)) to the kernel/regular network is path-consistent after arc-consistency enforcement.

The off-chain extension network has EQ diagonals for all elements (including the unfolded chain elements, which have replaced the kernel's PP self-loops). Therefore Theorem 2.3 (full RCC5 tractability) applies directly: this path-consistent disjunctive network is **satisfiable**, and a consistent atomic refinement exists. Choose such a refinement. Each off-chain witness w receives specific atomic relations $\rho(w, e)$ to all other nodes.

Moreover, because every value in the refined domains was in $\mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$, the chosen atomic relation $\rho(w, e)$ respects all universal constraints between w and e.

Step 5: Demand satisfaction.

- **$\exists\mathsf{PP}$ demands within chains:** For each τ_i in descriptor Desc_α and each $\exists\mathsf{PP}.C \in \tau_i$: if $C \in \tau_j$ for some j (within-period witness), then the element at the next occurrence of τ_j on the chain witnesses the demand ($\rho = \mathsf{PP}$ by transitivity). If C is not in any period type, the demand is witnessed by an off-chain PP-witness (from V6).
- **Non-PP demands of chain elements:** For $\exists R.C \in \tau_i$ (a phase type of Desc_α) with $R \neq \mathsf{PP}$: since $\exists R.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$, the quotient provides a designated witness w with $C \in \mathsf{tp}(w)$ and $\rho(k_\alpha, w) = R$ (T2, item 4). The chain element $d_j^{(\alpha)}$ of type τ_i inherits $\rho(d_j^{(\alpha)}, w) = \rho(k_\alpha, w) = R$ (Step 2), so w witnesses the demand.
- **Demands of regular nodes:** The quotient provides designated witnesses (T2, item 5).
- **Universal constraints:** For $\forall R.C \in \tau_i$:
  - Within chains ($R \in \{\mathsf{PP}, \mathsf{PPI}\}$): follows from (V2) for $R = \mathsf{PP}$ and (V3) for $R = \mathsf{PPI}$.
  - Within chains ($R \in \{\mathsf{DR}, \mathsf{PO}\}$): vacuously satisfied — no chain element is DR or PO from another element on the same chain (they are all PP/PPI-related).
  - Cross-chain and off-chain: if $\rho(d_j^{(\alpha)}, e) = R$ for an external element e, then $R \in \mathsf{Safe}(\tau_i, \mathsf{tp}(e))$ is guaranteed. For designated witnesses, this is part of (T2, item 4). For off-chain witnesses, this follows from the type-safety filter in (1): the atomic refinement chosen in Step 4 selects only type-safe relations.

Step 6: Global consistency. The full atomic network (chains + off-chain witnesses + regular nodes) is composition-consistent:

- Within-chain triples: Step 3.
- Triples involving one chain element and two kernel/regular/off-chain nodes: equivalent to the corresponding kernel triple (by stabilization) or covered by the V6 solution.
- Triples of off-chain witnesses: consistent by the atomic refinement from Step 4.

By the patchwork property, this composition-consistent atomic network is globally consistent and realizable. The realization, combined with the type assignments and the verified demand satisfaction (Step 5), gives a valid $\mathcal{ALCI}_{RCC5}$ model $\mathcal{I}' \models C_0$. □

9 Main result: decidability

Theorem 9.1 (Decidability of $\mathcal{ALCI}_{RCC5}$)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable.*

Proof. Given C_0:

1. Compute the bound $f(|C_0|)$ on quotient size (Theorem 7.1).
2. Enumerate all two-tier quotient structures of size $\leq f(|C_0|)$. This is a finite set: finitely many period descriptors (sequences of types of bounded length), finitely many (descriptor, external interface) pairs for kernels, finitely many regular configurations, finitely many atomic RCC5 labelings.
3. For each quotient structure, check validity: (V1)–(V4) for each descriptor, (V5) composition consistency of the kernel/regular network, (V6) path-consistency of the type-safety-filtered off-chain extension. Each check is decidable (finite computation).
4. Accept iff any valid quotient is found.

Correctness:

- *No false negatives:* If C_0 is satisfiable, then by Theorem 7.1, a valid quotient of size $\leq f(|C_0|)$ exists, so the procedure accepts.
- *No false positives:* If a valid quotient exists, then by Theorem 8.1, C_0 is satisfiable.

The enumeration is finite and each check is decidable, so the procedure terminates. □

10 Complexity

The quotient size bound $f(|C_0|)$ is large:

- Period descriptors: length $\leq |\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$. Number of distinct descriptors $D \leq 2^{2^{|\mathsf{cl}(C_0)|}}$.
- Kernel nodes: one per (descriptor, external interface) pair. The number of distinct interfaces per descriptor is at most 4^{K+L}. Since K and L are bounded (below), this is finite but involves a tower of exponentials.

- Regular nodes: bounded by (blocking classes) $\times$ (tree depth). The number of blocking classes is $|\mathsf{Tp}(C_0)| \times 4 \times 4^K$ where K is the number of kernels.
- The V6 check (arc-consistency with type-safety filtering) runs in polynomial time in the network size.

The decision procedure is therefore computable but with a very high complexity bound. We do not claim ExpTime membership.

Remark 10.1 (Towards tighter bounds)**.** The doubly-exponential bound on descriptors is likely a gross overcount. In practice, the number of distinct PP-chain behaviors is much smaller. An ExpTime upper bound would require showing that the quotient size is singly exponential, which we leave as an open problem. The lower bound is ExpTime-hard (inherited from $\mathcal{ALCI}$).

11 Discussion

11.1 The role of full RCC5 tractability

The soundness proof (Theorem 8.1) relies essentially on Theorem 2.3: the full tractability of RCC5. This deep result from qualitative spatial reasoning—that path-consistency decides satisfiability for the entire RCC5 algebra—is what bridges the gap between the quotient's local consistency (V5, V6) and the model's global consistency.

Without full tractability, we would need to verify global consistency directly, which is the essence of the "extension gap" in earlier approaches. Full tractability collapses the gap: local consistency (checkable) implies global consistency (needed).

11.2 Why earlier approaches had gaps

Approach	Gap	Resolution in this paper
Quasimodel [2]	Extension gap: Henkin construction may fail	V6 + full tractability: path-consistent extensions always succeed
Contextual tableau [4]	FW(C, N) false: infinite chains can't fit in fixed-width states	Period descriptors: chains as words, not graphs
Triangle types [7]	Extension solvability conjecture	V6 is extractable from models (model's assignment survives AC)

11.3 Why the V6 argument works where Q3s failed

The quasimodel approach required Q3s—arc-consistency on abstract *type-level* disjunctive networks. Computational verification showed that 11.1% of model-derived type-level networks violate Q3s. The key difference: V6 operates on *instance-level* networks with specific atomic relations between existing nodes, not on abstract type-level domains. The model provides actual relations for each witness that satisfy *all* composition constraints simultaneously, so arc-consistency cannot eliminate them.

11.4 Type-safety and singleton compositions

The RCC5 composition table has exactly four singleton entries: $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$, $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$. In these cases, the composition domain contains a single value R. If the type-safety filter $\mathsf{Safe}(\tau_w, \tau_e)$ excludes R, the V6 domain becomes empty.

This cannot happen for quotients extracted from models: the unique value R is the model's actual relation, which is always type-safe (Lemma 2.2). But it *does* mean that V6 with the type-safety filter is strictly stronger than V6 without it: some abstract quotients that pass the composition-only check will fail the type-safety-filtered check. This is correct behavior—those quotients would have produced unsound models.

11.5 Extension to $\mathcal{ALCI}_{RCC8}$

The argument relies on two properties of RCC5:

1. Full tractability (Theorem 2.3).
2. Universal self-absorption of PP (Theorem 3.1).

For RCC8, PP splits into TPP (tangential) and NTPP (non-tangential). NTPP is transitive ($\mathsf{comp}(\mathsf{NTPP}, \mathsf{NTPP}) = \{\mathsf{NTPP}\}$), but RCC8 is *not* fully tractable—only specific subalgebras are tractable [9]. The decidability of $\mathcal{ALCI}_{RCC8}$ remains open and may require additional techniques.

12 Conclusion

We have proved that concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable. The proof constructs a finite two-tier quotient for any satisfiable concept: period descriptors capture the periodic structure of infinite PP-chains (Tier 1), and PP-kernel nodes indexed by (descriptor, external interface) pairs capture cross-chain interactions (Tier 2). Completeness follows from the T_∞ characterization of recurrent chain types and the model's global consistency (which ensures the type-safety-filtered extension condition V6). Soundness follows from V6 and the full tractability of RCC5 (path-consistent networks are satisfiable), with the type-safety filter ensuring that the resulting atomic refinement respects all universal constraints.

The decidability of $\mathcal{ALCI}_{RCC5}$ has been an open problem since Wessel [1]. Our resolution relies on three key ingredients: (1) the algebraic self-absorption of PP in the RCC5 composition table, (2) the patchwork property of RCC5 from qualitative spatial reasoning, and (3) a type-safety filter ensuring that RCC5-consistent assignments are also DL-consistent.

The exact complexity remains open. The decision procedure's upper bound is computable but not elementary. An ExpTime upper bound (matching the $\mathcal{ALCI}$ lower bound) would require tighter analysis of the quotient size.

The decidability of $\mathcal{ALCI}_{RCC8}$ remains open, as RCC8 lacks full tractability.

References

[1] Michael Wessel. *Qualitative Spatial Reasoning with the ALCIRCC Family – First Results and Unanswered Questions.* Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.

[2] Claude (Anthropic). *On the Decidability of ALCIRCC5 and ALCIRCC8: Quasimodels Meet the Patchwork Property.* Discussion draft, March 2026.

[3] Claude (Anthropic). *Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction* (original version). Discussion draft, April 2026.

[4] GPT-5.4 Pro. *A Contextual Tableau Calculus for $\mathcal{ALCI}_{RCC5}$.* Discussion draft, 2026.

[5] GPT-5.4 Pro. *Technical Response to "Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction".* Review note, April 2026.

[6] Claude (Anthropic). *On the Failure of Finite-Width Extraction for $\mathcal{ALCI}_{RCC5}$.* Discussion note, March 2026.

[7] Claude (Anthropic). *Triangle Types and the Extension Gap in $\mathcal{ALCI}_{RCC5}$.* Discussion draft, March 2026.

[8] Jochen Renz and Bernhard Nebel. On the complexity of qualitative spatial reasoning: A maximal tractable fragment of the Region Connection Calculus. *Artificial Intelligence*, 108(1–2):69–123, 1999.

[9] Jochen Renz. Maximal tractable fragments of the Region Connection Calculus: A complete analysis. In *Proc. IJCAI*, pages 448–454, 1999.

[10] David A. Randell, Zhan Cui, and Anthony G. Cohn. A spatial logic based on regions and connection. In *Proc. KR*, pages 165–176, 1992.